

# RISHI KIRAN KARNATAKAM

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## SUMMARY

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Motivated Computer Science student seeking job and internship opportunities to apply my skills in software development and problem-solving. Quick to learn new technologies and adapt to challenges, focusing on contributing to innovative and impactful projects.

## EDUCATION

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| <b>NNRESGI</b>                                                             | Dec 2021 - Present |
| <i>Bachelor of Technology in Computer Science and Engineering GPA: 8.4</i> |                    |

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|---------------------------------------|---------------------|
| <b>KMIIT</b>                          | Sep 2019 - Jun 2021 |
| <i>Intermediate (10+2 equivalent)</i> | <i>GPA: 9.4</i>     |

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| <b>SAGHS</b>                        | May 2019        |
| <i>Secondary School Certificate</i> | <i>GPA: 9.8</i> |

## TECHNICAL SKILLS

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**Programming Languages:** C, Python, Dart, C#

**Databases:** MySQL, MongoDB

**Cloud & DevOps:** Amazon Web Services (AWS), Git, Github, Jenkins

**Frameworks & Libraries:** TensorFlow, Flask, Tkinter, Flutter, Unity Engine

## PROJECTS

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|----------------------------------------------------------------------------------|----------|
| <b>Cotton Plant Disease Detection and Classification Using Transfer Learning</b> | Feb 2024 |
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Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.

- Improved disease detection accuracy by 25% using transfer learning with pre-trained CNN models.
- Achieved 90%+ accuracy on test datasets, optimizing performance for agricultural disease diagnosis.

### **AI-Powered Chess Engine with Genetic Algorithm Optimization** Aug 2024

Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.

- Enhanced strategic depth and move accuracy by 20%.
- Achieved a 30% improvement in move generation speed through alpha-beta pruning.

### **AI-Integrated Plant Disease Detection Mobile App** Apr 2025

Built a mobile app for real-time plant disease detection using camera or gallery images.

- Integrated a TensorFlow Lite model (Inception V3) for accurate disease prediction with offline support.
- Enabled local data storage and automatic syncing to MongoDB Atlas when connected online.
- Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.

### **Interactive Solar System Model and 3D Game Development** Oct 2023

Developed a 3D interactive solar system model to enhance gamified learning.

- Created a first-person 3D game inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.

## **AWARDS**

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**Internal Hackathon on Generative AI** (Aug 2024): Achieved 3rd place in a college-hosted Hackathon focused on Generative AI.

**National Hackathon on Generative AI** (Sep 2024): Achieved 2nd place in a National Level Hackathon focused on Generative AI.

**National Hackathon on AR/VR Development** (Oct 2023): Ranked in the top 10 at a national-level hackathon focused on AR/VR.

## **CERTIFICATIONS**

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**AWS Academy Graduate - AWS Academy Cloud Architecting** (Sep 2024) - AWS Academy

**AICTE Virtual Internship** (2024) - Nalla Narasimha Reddy Education Society's Group of Institutions

**Python For Data Science** (Jan 2025) - IIT Madras (NPTEL)

**The Joy of Computing Using Python** (Jul 2023) - IIT Madras (NPTEL)

**Supervised Machine Learning: Regression and Classification** (Jun 2023) - Stanford University (Coursera)